Answer the following questions:

1. Write a short note on anamolous properties of the first element of p-block.
2. Describe briefly allotropism in p- block elements with specific reference to carbon.
3. Boron does not react directly with hydrogen. Suggest one method to prepare diborane from BF_3.
4. Give the uses of Borax.
5. What is catenation ? describe briefly the catenation property of carbon.
6. Write a note on Fisher tropsch synthesis.
7. Give the structure of CO and CO_2.
8. Give the uses of silicones.
9. $AlCl_3$ behaves like a lewis acid. Substantiate this statement.
10. Describe the structure of diborane.
11. Write a short note on hydroboration.
12. Give one example for each of the following

    (i) icosogens (ii) tetragen (iii) prictogen (iv) chalcogen

13. Write a note on metallic nature of p-block elements.
14. Complete the following reactions

    a. $B(OH)_3 + NH_3 \longrightarrow$

    b. $Na_2B_4O_7 + H_2SO_4 + H_2O \longrightarrow$

    c. $B_2H_6 + 2NaOH + 2H_2O \longrightarrow$

    d. $B_2H_6 + CH_3OH \longrightarrow$

    e. $BF_3 + 9\ H_2O \longrightarrow$

    f. $HCOOH + H_2SO_4 \longrightarrow$

    g. $SiCl_4 + NH_3 \longrightarrow$

    h. $SiCl_4 + C_2H_5OH \longrightarrow$

    i. $B + NaOH \longrightarrow$

    j. $H_2B_4O_7 \xrightarrow{\text{Red hot}}$
15. How will you identify borate radical?
16. Write a note on zeolites.
17. How will you convert boric acid to boron nitride?
18. A hydride of 2nd period alkali metal (A) on reaction with compound of Boron (B) to give a reducing agent (C). identify A , B and C.
19. A double salt which contains fourth period alkali metal (A) on heating at 500K gives (B). aqueous solution of (B) gives white precipitate with $BaCl_2$ and gives a red colour compound with alizarin. Identify A and B.
20. CO is a reducing agent . justify with an example.

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